

Pharos

A physics-grounded mixture-of-experts architecture
for RNA three-dimensional structure prediction

Technical report — the architecture as built, and what it measures

SBS-RNA Project

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Abstract

This report documents PHAROS as it exists in code, not as it was designed. It is a 395.3M-parameter model (306.0M active) that predicts RNA secondary structure, chemical reactivity, contacts, distances, Leontis–Westhof pair geometry, Mg^{2+} sites, crystallographic rigidity, backbone pseudotorsions and three-dimensional backbone coordinates, trained through a six-stage curriculum from a 41.2M-sequence masked-language-modelling corpus down to a 16,604-chain structural set.

We report what it achieves, which is a mean TM-score of **0.094** over 17 RNA-Puzzles targets against a field median of 0.301 — last place on 15 of 17 — alongside a contact average precision of 0.890 and an lDDT of 0.269. That combination is the central empirical fact of this report: **the model builds locally-valid RNA backbone and places it in the wrong global arrangement**. We trace it to a specific architectural omission, give the fix, and are explicit that the fix is not yet measured.

The second contribution is methodological. The project maintains a numbered defect register, at present 104 entries, whose recurring class is *a component that exists, is measured, and whose output is never checked for basic validity*. Several entries are errors in our own instrumentation rather than in the model, and are recorded as such. We argue this bookkeeping is not incidental to the work but is the main reason the negative results above can be trusted.

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1 What problem this is

RNA structure prediction has not had its AlphaFold2 moment. On non-redundant benchmarks the best current methods reach mean TM-scores around 0.5–0.6, where 0.45 is roughly the threshold for having recovered the correct fold at all, against routine scores above 0.9 for proteins. The usual explanations are that there are two orders of magnitude fewer RNA structures than protein structures, that RNA’s functional unit is often a conformational ensemble rather than one structure, and that RNA folding is ion-dependent in a way protein folding is not.

PHAROS takes a position on which of these matters: that the binding constraint is *inductive bias* rather than scale, and that the physics of ion condensation and the geometry of base pairing should enter the architecture rather than be left for the network to rediscover. This report does not defend that position — it was argued in the design report that preceded the implementation — but documents what was built from it and what that produced.

A note on what this report is not. It reports a model that does not work yet at the task it was built for. The results section is written to be useful to that end rather than flattering, and every number in it is traceable to a file in the repository.

2 Architecture

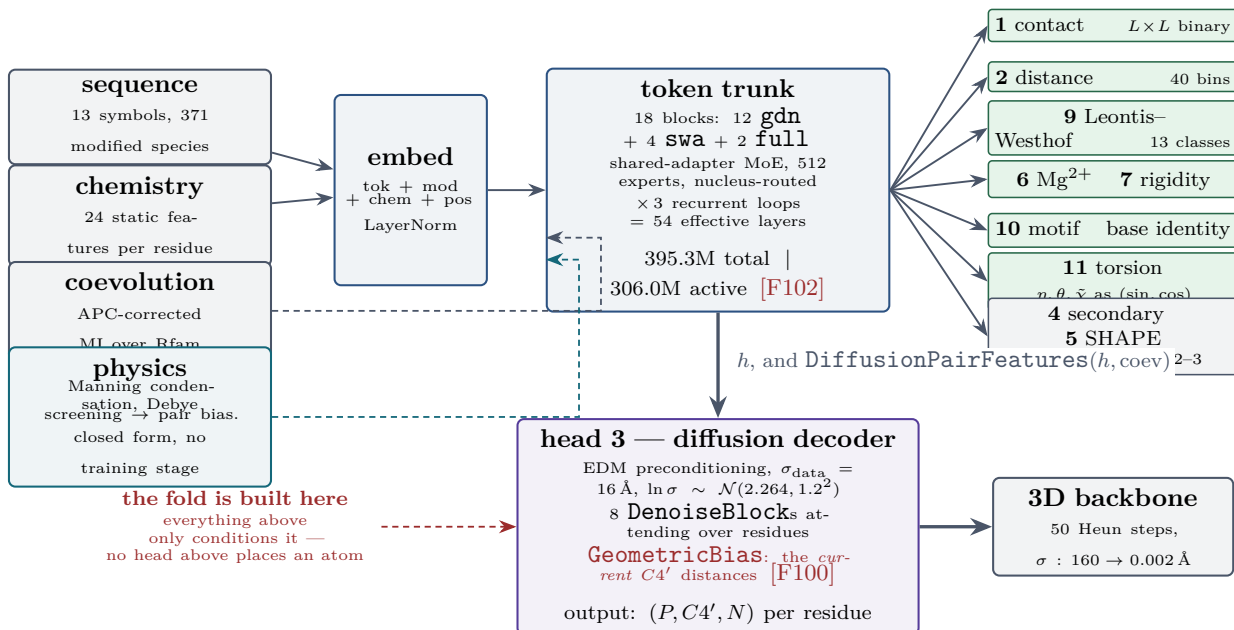


Figure 1: Pharos end to end. A shared trunk produces one residue representation h that every head reads. Ten heads are trained discriminative readouts; the eleventh, head 3, is a diffusion decoder that is the only component that places an atom. The physics module contributes a closed-form electrostatic pair bias and has no training stage of its own. Bracketed numbers are defect-register entries discussed in §6.

2.1 Inputs

Each residue carries four things, summed and layer-normalised:

- a **symbol** from a 13-token vocabulary (A C G U N DA DC DG DT DU I UNK PAD), where N means the ribose is modelled but the base identity is unassigned;

- a **modification** id over 371 species (index 0 is NO_MOD), derived from the chemical component dictionary, so that a pseudouridine is not silently read as a uridine;
- 24 **chemistry** features through a single affine map — a 5-way one-hot, a purine flag, six per-edge hydrogen-bond donor/acceptor counts (Watson–Crick, Hoogsteen/CH, sugar), the pK_a of the protonating ring nitrogen and a shifted- pK_a flag, two sugar pucker propensities, polarisability and a stacking-energy prior, the phosphate charge state, a 2'-OH flag, three modification-class indicators, and local GC fraction in a ± 16 window;
- a **learned absolute position** embedding over 4,608 slots. There is no rotary or sinusoidal encoding anywhere in the model.

2.2 The trunk

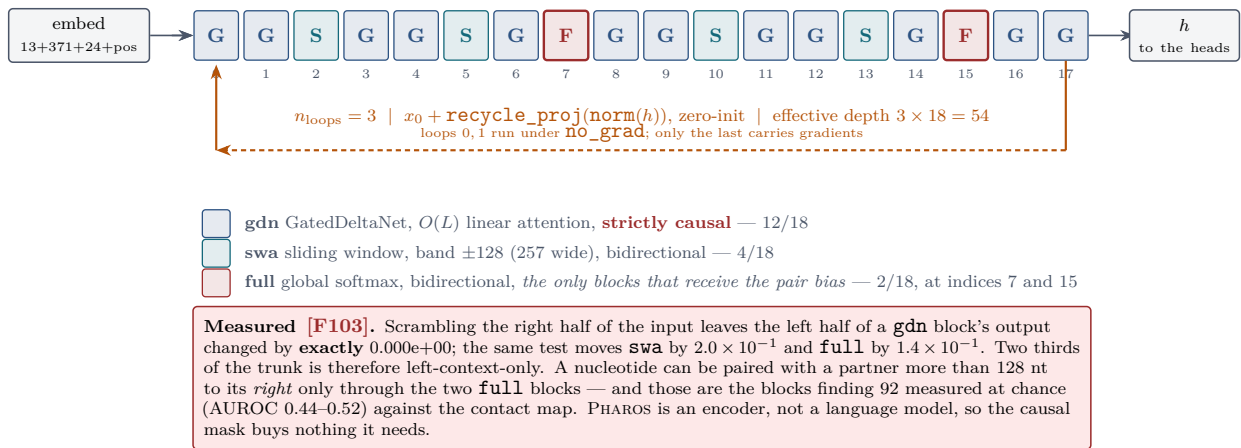


Figure 2: The 18-block trunk. A period-8 attention pattern tiled to 18 blocks, run three times with the previous pass recycled in. The annotation records a property we measured rather than assumed, and which we regard as an open question rather than a settled defect.

The trunk is 18 pre-norm blocks following the period-8 pattern (**gdn**, **gdn**, **swa**, **gdn**, **gdn**, **swa**, **gdn**, **full**), which tiles to twelve **gdn**, four **swa** and two **full** at indices 7 and 15. Only the **full** blocks receive the electrostatic pair bias, through a per-head gate initialised to zero.

gdn — gated delta networks. Linear attention with a delta rule, $O(L)$ in sequence length. Per head, with state $S \in \mathbb{R}^{d_k \times d_v}$,

$$S_t = a_t S_{t-1} \left(I - b_t k_t k_t^\top \right) + b_t v_t k_t^\top, \quad o_t = S_t^\top q_t, \quad (1)$$

with $a_t, b_t \in (0, 1)$ sigmoid gates, q and k ℓ_2 -normalised, and an output gate on o . The delta term erases the component of the state already associated with k before writing, so a recurring motif overwrites rather than accumulates — on a helix, where the same local geometry recurs every ~ 11 nt, accumulation is precisely the failure mode. It is evaluated chunkwise in blocks of 128 with a triangular solve.

swa and full. Sliding-window attention over a band of ± 128 (257 wide), implemented densely with a band mask, and ordinary global softmax attention.

Recurrence. The entire trunk is run $n = 3$ times with shared weights, giving 54 effective layers. Loops 0 and 1 run under `no_grad` and only the last carries gradients, so the memory cost is that of one pass. The recycled state enters as $x_0 + \text{recycle_proj}(\text{norm}(h))$ with `recycle_proj` initialised to zero, so loop 1 begins numerically identical to loop 0. Training samples the depth uniformly from $1 \dots 3$. There is no halting mechanism.

2.3 The mixture of experts

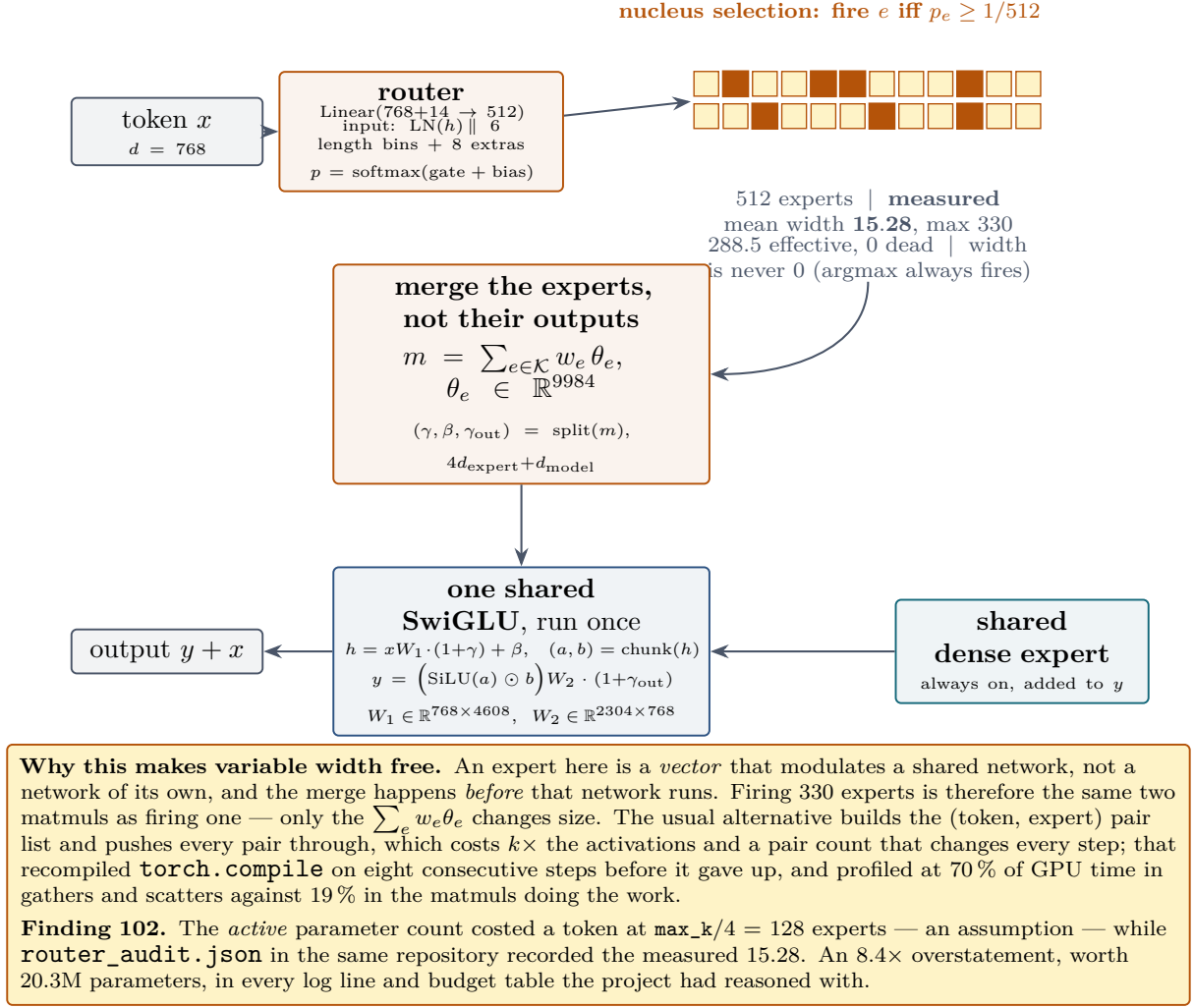


Figure 3: Shared-adapter MoE with nucleus routing. An expert is a vector that modulates one shared SwiGLU, not a network of its own, so firing 300 experts costs the same matmul as firing one. Routing width is variable and measured, not assumed — getting that wrong is finding 102.

Every block’s feed-forward is a mixture of 512 experts plus one always-on dense expert. Two choices distinguish it from the usual top- k arrangement.

Experts are vectors, not networks. One shared SwiGLU $W_1 \in \mathbb{R}^{768 \times 4608}$, $W_2 \in \mathbb{R}^{2304 \times 768}$ is modulated per token by a convex combination of the selected experts’ parameter vectors:

$$m = \sum_e w_e \theta_e, \quad (\gamma, \beta, \gamma_{\text{out}}) = \text{split}(m), \quad y = (\text{SiLU}(a) \odot b)W_2 \cdot (1 + \gamma_{\text{out}}), \quad (2)$$

where $h = xW_1 \cdot (1 + \gamma) + \beta$ and $(a, b) = \text{chunk}(h)$. Each θ_e has $4d_{\text{expert}} + d_{\text{model}} = 9,984$ parameters. Because the merge happens *before* the network runs, a variable number of active experts costs no extra matmul — which is what makes the next choice affordable.

Nucleus routing, not top- k . An expert fires when its softmax probability clears $1/E$, i.e. when the token prefers it to chance, with the argmax always included so the width is never zero:

$$\mathcal{K}(x) = \{e : p_e \geq \tau/E\} \cup \{\arg \max_e p_e\}, \quad \tau = 1. \quad (3)$$

A relative threshold is scale-free in a way an absolute one is not: at initialisation every expert sits at $1/E$, so a fixed 0.02 fires all 32 experts of a small config and none of 256. The measured width

on a trained checkpoint is **15.28** experts on average with a maximum of 330, 288.5 effective experts and no dead expert.

Load balance. With m_i the padding mask, f_{ie} the fired indicator and $\bar{k}$ the mean width, $\text{bal} = E \sum_e \text{frac}_e \cdot \bar{p}_e$ summed over blocks at weight 0.01, giving a floor of $0.01 \times 18 = 0.18$ at perfect balance.

2.4 Parameter budget

	shared400	base400	small	mini
d_{model}	768	640	512	384
blocks $\times$ loops	18×3	18×8	16×8	12×12
heads	12	10	8	6
d_{pair}	160	160	128	128
experts $\times d_{\text{expert}}$	512×2304	48×192	32×256	32×192
routing	nucleus	top-6	top-4	top-4
total parameters	395,295,754	405,924,150	259,188,898	113,658,264
active parameters	306,001,570	127,232,310	83,028,130	39,340,440

Table 1: Configurations. For **shared400**, 3.85M of the total is embeddings, 340.8M the trunk (187.6M of it experts) and 48.6M the heads. The active count costs a token at the *measured* routing width of 15.28; it previously used $\text{max_k}/4 = 128$ and overstated the figure by 20.3M (finding 102).

3 The diffusion decoder

Head 3 is the only part of PHAROS that places an atom. It denoises three backbone atoms per residue — P, C4' and the glycosidic nitrogen, the minimum that fixes a frame — under EDM preconditioning:

$$D(x; \sigma) = c_{\text{skip}}(\sigma) x + c_{\text{out}}(\sigma) F(c_{\text{in}}(\sigma) x, c_{\text{noise}}), \quad c_{\text{skip}} = \frac{\sigma_d^2}{\sigma^2 + \sigma_d^2}. \quad (4)$$

with $\sigma_d = 16 \text{ \AA}$ set from the corpus rather than inherited, and $\ln \sigma \sim \mathcal{N}(p_{\text{mean}}, 1.2^2)$ where $p_{\text{mean}} = \ln(0.6016 \sigma_d) = 2.2644$ is *derived* from σ_d and never typed beside it. The derivation matters: Karras et al. recommend $P_{\text{mean}} = -1.2$ for $\sigma_d = 0.5$, which is a median σ/σ_d of 0.602; carrying the -1.2 over while raising σ_d to 16 for ångströms puts the median draw at $c_{\text{skip}} = 0.99965$, where the network is asked to supply 0.035% of the answer (finding 85).

The loss is the σ -weighted denoising error plus a flat-bottomed geometry penalty at weight 0.1:

$$\mathcal{L} = \mathbb{E} \left[\frac{\sigma^2 + \sigma_d^2}{(\sigma \sigma_d)^2} \|D(x_0 + \sigma \epsilon; \sigma) - x_0\|^2 \right] + 0.1(v_{\text{C4'-N}} + v_{\text{P-P}}), \quad (5)$$

with $v = \max(0, |d - d_0| - \text{tol})$ for $d_0 = 3.38 \text{ \AA}$ (tol 0.25) and $d_0 = 6.01 \text{ \AA}$ (tol 1.50). Sampling is 50 Heun steps on a $\rho = 7$ schedule from $\sigma_{\text{max}} = 160$ to $\sigma_{\text{min}} = 0.002$.

3.1 What the decoder could not do

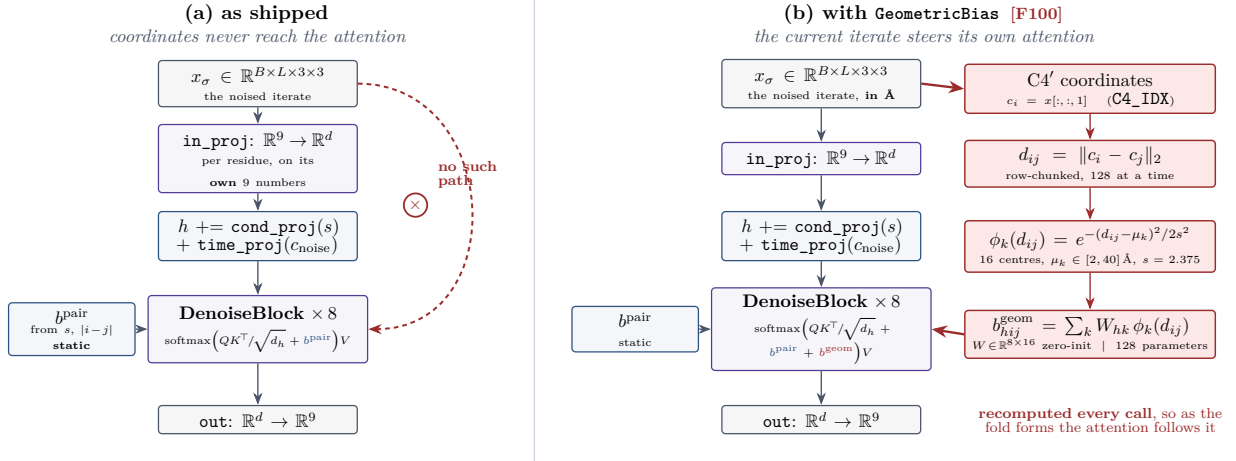


Figure 4: Finding 100. The decoder had no mechanism by which one residue’s position could influence another’s update. Coordinates entered through a per-residue linear map of that residue’s own nine numbers, and the attention bias was built once from hidden states and relative *sequence* position. Two residues 6 Å apart in space were, to this attention, exactly as unrelated as two 60 Å apart.

This is the architectural claim of the report, so it is worth stating carefully. In the shipped decoder,

$$h \leftarrow \text{in_proj}(\text{vec}(x_\sigma)_i) + \text{cond_proj}(s_i) + \text{time_proj}(c_{\text{noise}}), \quad h \leftarrow \text{Block}(h, b^{\text{pair}}), \quad (6)$$

where $\text{vec}(x_\sigma)_i$ is residue i ’s own nine coordinates and b^{pair} is a function of s and $|i-j|$ only. Neither term couples residue i to *where residue j currently is*.

This predicts exactly the failure we observe. Bonds come out right, because relative sequence position suffices for a bond. The fold comes out wrong, because nothing couples distant residues by proximity. And feeding the decoder a predicted contact map changed nothing (finding 93, +0.0025 TM against a 0.0023 noise floor, sign test $p = 0.50$) — because a static map is not current geometry: to decide how to move an atom the model must know where it is *now*.

The fix embeds the pairwise C4’ distances of the current iterate in a 16-centre radial basis over 2–40 Å and projects to a per-head additive attention bias,

$$b_{hij}^{\text{geom}} = \sum_k W_{hk} \exp\left(-\frac{(d_{ij} - \mu_k)^2}{2s^2}\right), \quad d_{ij} = \|c_i - c_j\|_2, \quad (7)$$

recomputed on every call so the attention follows the structure as it forms. W is zero-initialised, so a loaded checkpoint’s forward pass is bit-identical until the bias earns its way in. The whole cost is **128 parameters**.

Properties pinned by test: zero at initialisation; symmetric; SE(3) invariant to 2.03×10^{-6} , which matters because the loss augments with random rotations; and — the property that was absent — moving one residue changes the predicted update for *the others*.

Two defects in the fix itself. The first version was fed x_σ , which is $c_{\text{in}}x$ and not ångströms. At the median σ that divides every distance by 18.7, putting the entire informative range below the first basis centre and cutting summed per-channel resolving power from 11.42 to **0.72**. It would have trained to nothing while all six tests passed, because all six called the module directly with ångströms — the module was right and its caller was wrong. The second built its basis in fp32 inside a bf16 autocast region and cost twice the activation memory it needed, which turned a run that had always fitted into a fatal out-of-memory at epoch 25 of 40. Both are in the register as 100b and 104b.

4 Heads, losses and curriculum

#	head	predicts	loss	weight
1	contact	heavy-atom contact $< 8 \text{ \AA}, i - j \geq 4$	BCE, quality-weighted	1.0
2	distance	C4'–C4', 40 bins	cross-entropy	0.5
3	structure	backbone (P, C4', N)	EDM + violation	1.0
4	secondary	dot-bracket, 8 symbols	cross-entropy	0.5
5	reactivity	SHAPE and DMS	smooth- L_1	1.0
6	Mg ²⁺	ion within 3 Å	BCE, clamped pos-weight	0.3
7	rigidity	z -scored B-factor, X-ray only	smooth- L_1	0.3
8	disorder	unobserved residue	none	—
9	geometry	Leontis–Westhof class, 13	cross-entropy	0.5
10	motif	hairpin / internal / neither	cross-entropy	0.3
11	torsion	$\eta, \theta, \tilde{\chi}$ as (sin, cos)	unit-circle + norm	0.3
–	base identity	4 classes	cross-entropy	0.2
–	fluctuation	ensemble amplitude	smooth- L_1	0.1
–	fitness	variant fitness	MSE	0.3
–	splice, ensemble state	—	none	—

Table 2: The head inventory. Three outputs are emitted and never trained: disorder needs `entity_poly_seq` in the corpus, splice needs a corpus that has not been acquired, and no observable distinguishes the K ensemble states. They are recorded as untrained rather than quietly reported.

Torsions. Head 11 predicts the Duarte–Pyle pseudotorsions $\eta_i = \angle(C4'_{i-1}, P_i, C4'_i, P_{i+1})$, $\theta_i = \angle(P_i, C4'_i, P_{i+1}, C4'_{i+1})$ and $\tilde{\chi}_i = \angle(P_i, C4'_i, N_i, P_{i+1})$ as unnormalised (sin, cos) pairs, with loss $\|u - t\|^2 = 2 - 2 \cos \Delta$ on the normalised vector plus $0.02 \|p\| - 1$ on its magnitude. They are computed from batch coordinates on the fly, never stored.

Distance bins. 40 bins: eight uniform of 0.875 Å over 3–10 Å, then 32 equal-frequency edges to 100.78 Å with the last open-ended. Uniform binning over the whole range had put the majority class at a level the head could reach by constant prediction, which is finding 75.

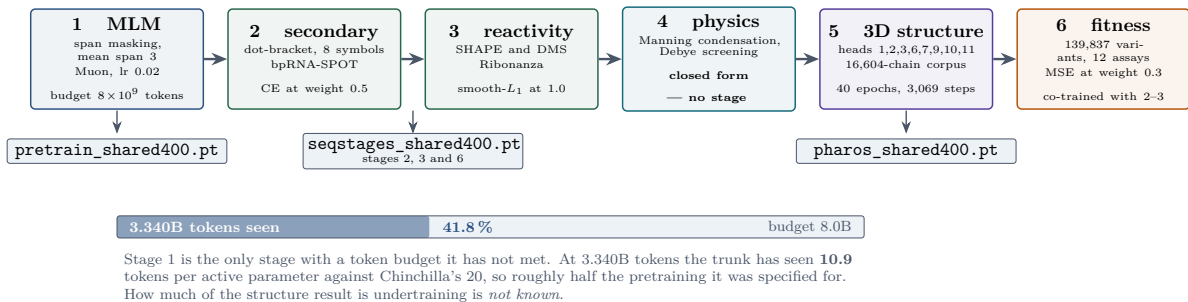


Figure 5: The six-stage curriculum. Each stage initialises from the previous one’s checkpoint, and the driver refuses to start a stage from random weights.

5 Results

5.1 What the heads learn

At the end of a 40-epoch stage-5 run (3,069 optimiser steps, 16,384-token budget) on the held-out test split:

metric	value	reference
contact average precision	0.8901	base rate 0.4541 [F107]
Mg ²⁺ average precision	0.4167	base rate 0.0254 (16.4×)
Leontis–Westhof accuracy	0.8972	majority 0.8532
motif accuracy	0.7494	majority 0.6925
η MAE	23.29°	circular baseline 24.59°
θ MAE	26.86°	circular baseline 29.05°
$\tilde{\chi}$ MAE	16.21°	circular baseline 17.74°
structure MSE	24.59	—
P–P bond violation	0.1934	target 6.01 Å

Torsions beat their circular baselines by 1.3–2.2°, which is real but small. Ion sites are 16× better than chance.

The contact number needs a caveat, and it is a large one. That average precision is computed on pairs from `sample_pairs`, which takes *every* positive plus at most 512 sampled negatives, so the set it scores is **45.4% positive**. The task — every pair with $|i - j| \geq 4$ — is **9.6% positive** on this corpus, which we measured directly over 200 test chains. Average precision moves with the base rate, so 0.8901 at a 0.4541 base is not comparable to a published contact-prediction figure, and the honest normalised form is the lift the code also reports: 1.96× base. The full-pair-set metric is implemented and has not yet been run, because it needs a GPU (finding 107).

This matters beyond bookkeeping: 0.890 is the number §5 and finding 92 use to argue that contacts are predicted well while the fold is not, and that argument is what sent the search into the decoder. It may survive — see below — but it has not been tested on the task it claims.

One hypothesis, tested and refuted. The pair feature is an outer sum $As_i + Bs_j$, which can score well by knowing *which* residues are paired without knowing *with whom* — and that would have explained AP 0.89 beside TM 0.09 neatly. It does not. A partner-blind oracle handed the *true* degree of every residue reaches only AP 0.3005 on the real pair set, against a 0.0961 base and 0.1209 for sequence separation alone. The head sits far above that, so it does carry genuine pairing-partner information.

5.2 What the model does not do

	run 1	run 2	run 3	field median
mean TM-score	0.0890	0.0913	0.0936	0.3013
mean IDDT	0.2550	0.2765	0.2690	—
targets above field median	0/17	0/17	0/17	—
targets above TM 0.45	0/17	0/17	0/17	—

Table 3: 17 RNA-Puzzles targets, scored against all submitted competitor models. Three runs at identical settings span 0.0046 against a 0.0023 initialisation-noise floor measured between two of them. Nothing has moved.

The mean percentile against the field is 99.7%: last or near-last on 15 of 17 targets.

The diagnostic. IDDT 0.269 beside TM 0.094 is the whole story. IDDT is a local measure — it asks whether each residue’s neighbourhood is right — and TM is a global one. A model that scored badly on both would be producing noise. A model that scores *moderately* on the local measure and at floor on the global one is producing locally-valid RNA backbone in the wrong global arrangement. That is consistent with every other number here: bonds within tolerance, contacts at AP 0.890, torsions beating their baselines, and a fold that is wrong.

What has been ruled out. Chain breaks: 99.09 % of 2,227,344 P–P pairs lie inside tolerance. Stage-1 representation quality: sequence explains 0.92% of P–P distance variance, so the bond is a geometric constant and not something the language model must supply. Decoder access to contacts: supplying a predicted contact map moved TM by +0.0025 against a 0.0023 noise floor, 9 wins in 17, sign test $p = 0.50$ (finding 93, reverted).

What that leaves is the decoder’s ability to *use* geometry, which is §3 and finding 100. At the time of writing that A/B has completed 25 of 40 epochs and **no claim is made about its outcome**. At matched epochs the two arms are indistinguishable on structure loss; the bias itself is demonstrably learning, with per-head attention offsets that are positive at 6–12 Å and negative at 23–40 Å, which is the correct sign structure.

6 The defect register as method

The repository carries `research/architecture/FINDINGS.md`, at present 104 numbered entries. It is not a changelog. Its recurring class is narrow and worth naming:

a component that exists, is measured, and whose output is never checked for basic validity.

Representative entries:

- **85** — P_{mean} carried over from a paper that used $\sigma_d = 0.5$, leaving the denoiser responsible for 0.035% of its task at the median training draw.
- **90** — the inference sampler passed `pair=None`, withholding relative position from the decoder at sampling time only. P–P went from 16–19 Å to 5.53–6.43 Å against a true 5.95.
- **102** — the active parameter count costed a token at `max_k/4 = 128` experts while an audit file in the same repository recorded a measured width of 15.28. Both numbers were on disk; nothing compared them.
- **104** — the training loop skipped out-of-memory batches and the validation ten lines below it did not, so a batch that costs one gradient in training killed a 40-epoch run at epoch 25.

Three disciplines follow from this, and they are the reason the negative results in §5 can be relied on.

Every new pathway is zero-initialised. `bias_scale`, `site_scale`, `coev_proj`, `recycle_proj` and `GeometricBias.proj` all start at exactly zero, so adding a feature cannot regress a working checkpoint and the feature has to earn its contribution.

Every metric is reported against its floor. Accuracy against the majority class, average precision against the base rate, angular MAE against the circular mean, correlation pooled over the split rather than averaged per batch. Finding 75 was a head that looked like it was learning because its floor had never been computed.

Refutations are recorded as prominently as confirmations. The register contains a withdrawn finding (76, where we measured our own interference and blamed the code), a refuted experiment (93), a corrected claim (a +0.2761 distance lift computed against a floor sampled over different batches), and a section titled *Checked and not defects*. Twenty-two of a monitoring script’s first twenty-three warnings were false positives, and fixing the script was treated as equal in value to fixing the model.

7 Limitations, and what we think is wrong

1. **The fold is wrong and the fix is unmeasured.** Finding 100 has a mechanism that explains the observations and a bias that demonstrably learns the right sign structure, and that is all that can honestly be said until the A/B finishes.
2. **Two thirds of the trunk cannot see to its right.** `gdn` is strictly causal — scrambling the right half of the input moves the left half of its output by exactly zero. PHAROS is an encoder and nothing in the task asks for a causal mask. We record this as declared rather than as a defect, because contact AP of 0.890 shows the information does arrive, by two routes that do not need `gdn`: the pair features are an outer sum that sees both endpoints, and three recycled passes carry right-context back (finding 103).
3. **The trunk’s global attention does not attend to contacts.** The two full blocks score AUROC 0.44–0.52 against the contact map while the contact head reaches AP 0.890 — so contacts are being read off the pair features, not built by attention (finding 92, open).
4. **Three heads are emitted and untrained,** and one declared router feature is read by the model and written by no caller (finding 68, open).
5. **Stage 1 is at 41.8% of its token budget.** At 3.340B of 8B tokens the trunk has seen 10.9 tokens per active parameter, about half Chinchilla-optimal. Some of the gap is undertraining and we do not know how much.
6. **The fitness head barely transfers.** Validation Spearman +0.604 over 12 assays of seen constructs against +0.085 on a held-out aptamer family (finding 97, open).

The honest summary. PHAROS is a working multi-task RNA model with a non-functioning structure decoder. The discriminative heads do what they were built to do. The generative head does not, we believe we know why, and the experiment that would settle it is running.